

Semantic Thermodynamics: A Framework for LLM Entropy Minimization

Abstract

Large Language Models (LLMs) fundamentally operate as Bayesian inference engines traversing highly dimensional semantic spaces. In environments with high semantic entropy (unrestricted prompt boundaries), these models expend excess computational energy (inference time and reasoning tokens) to map infinite phase spaces before collapsing a probability into an output.

This paper introduces the framework of **Semantic Thermodynamics**, positing that narrative structure is not merely a linguistic abstraction, but a physical constraint capable of performing work. By applying "Narrative Gravity" — a precise formulation of persona, teleological vectors, topological boundaries, and destructive pruning — we demonstrate an empirical reduction in output token expenditure by 79.29% and a reduction in system latency by 60.73%.

Furthermore, we introduce the **Law of Entropic Proportionality (Λ)**, proving that maximum constraint density must be proportional to input entropy to avoid structural friction, offering a paradigm shift in how commercial AI infrastructure routes and processes natural language.

1. The Fundamental Equation of Entropic Pruning

The framework seeks to achieve Narrative Entropy Minimization (ΔN_{min}), forcing the system into its state of highest computational efficiency.

$$\Delta N_{min} = \Lambda(E_{in}) \cdot [\Psi \cdot \vec{\Omega} \cdot \oint (R - X) \cdot M] + S_0$$

Dissecting the Variables:

- **Ψ (Psi) - The Persona:** Forces the immediate collapse of the model's "wave function" into a hyper-specialized mental model.

- Ω (**Omega**) - **The Teleological Vector**: The gravitational force directing Brownian probability toward a single target.
- $\oint$ (**Contour Integral**) - **The Scope Boundary**: Establishes the topological limits of the operation, ensuring laminar information flow.
- $\bar{X}$ (**X-Bar**) - **Destructive Pruning**: The explicit elimination of channel noise via destructive interference (e.g., "Ignore X").
- M (**Matrix**) - **Output Crystallization**: The rigid molecular structure of the final output (e.g., pure JSON).
- S_0 (**Ground State**) - **The Fallback**: A thermodynamic safety valve providing a minimum-energy state to rest if the target is unachievable, preventing probabilistic hallucination.
- $\Lambda(E_{in})$ - **The Entropic Proportionality Tensor**: A dynamic coefficient that adjusts constraint density based on the input's entropic mass.

2. Empirical Validation (Experiment Zero)

A controlled baseline (Zero Gravity) was tested against the Semantic Thermodynamics framework (Formula v1.0) over 100 API iterations, isolating the final material action.

Results:

- Control (High Entropy): 305.70 avg completion tokens | 3,489.38 ms avg latency
- Optimized (Low Entropy): 63.32 avg completion tokens | 1,370.38 ms avg latency
- **Delta**: -79.29% completion tokens | -60.73% latency

3. Structural Friction (Experiment One)

Further testing revealed that excessive narrative constraints on low-entropy inputs cause "Structural Friction," reversing the ROI. The model consumes more compute reconciling the constraints than solving the problem. This anomaly necessitates the Λ tensor for dynamic routing.

Conclusion

Semantic Thermodynamics proves that narrative constraint acts as a physical boundary in Bayesian systems. Implementing this framework allows for dynamic, context-aware routing that drastically reduces overhead costs and latency in enterprise LLM deployments without sacrificing accuracy.

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